Iliass Sijelmassi

Paris, France | +33 6 48 74 04 04 | iliass.sijelmassi-idrissi@polytechnique.edu | iliasssijelmassi.com | github.com/iliassSjm

Summary

MSc Data Science & AI candidate (École Polytechnique & HEC Paris, GPA 3.8/4.0), Stanford visiting researcher applying ML to large-scale time-series prediction, and incoming Quantitative Researcher Intern at **WorldQuant**.

Education

École Polytechnique — MSc Data Science & AI (Year 1) GPA: 3.8/4.0 *Sep 2024 – Jun 2025*
- **Coursework:** Probability & Statistics, Linear Algebra & Regression, Optimization, Deep Learning, Machine Learning.

HEC Paris — MSc Data Science & AI (Year 2) *Sep 2025 – Jun 2026*
- **Quant-relevant coursework:** Machine Learning for Financial Markets, Time Series, Causal Inference, Reinforcement Learning, Interpretability & Algorithmic Fairness, ML Ops & Cloud Engineering.

INP–ENSEEIHT, Toulouse — Dipl. Eng. in CS, Telecom & Applied Math *Sep 2020 – Sep 2023*
- **Coursework:** Probability, Statistics, Optimization, RL, Algorithms, High-Performance Computing, Scientific Computing (**MATLAB**).

Experience

Quantitative Researcher Intern | ***WorldQuant** (**Millennium** affiliate), Paris* *Oct 2026 – Mar 2027*
- Incoming 6-month internship within a Millennium systematic pod; details confidential.

Visiting Student Researcher | ***Stanford University**, Stanford, CA* *Mar 2026 – Sep 2026*
- First-author study predicting a rare, severe post-surgery cardiac complication in **7,653** patients, combining structured medical records with high-frequency bedside **waveforms** (blood pressures, ECG) from a ˜**1.5 TB** archive; parsimonious pre-operative model reached **AUROC 0.74**, calibrated to the **1.4%** event rate.
- Designed a leakage-controlled evaluation across four model families (pre-operative data, early-ICU signals, missingness-aware and **FFT** waveform features), benchmarking logistic regression against gradient boosting, random forest and a convolutional baseline under patient-grouped cross-validation with in-fold feature selection; **externally validated on unseen data** from a second hospital system (**AUROC 0.695** on ˜11,800 admissions; **0.731** after recalibration).

Data Scientist Intern | ***Crédit Agricole CIB (CACIB)**, Paris* *Mar 2025 – Sep 2025*
- Modeled whether internal audit recommendations would be implemented on time across **200k+** historical records, reaching **F1 ≈ 0.75** and improving **ROC-AUC** over the heuristic baseline.
- Designed a dual-model approach — **regularized Logistic Regression** for the coefficient-level interpretability auditors required, plus **XGBoost** for performance; drove feature selection with domain experts and validated robustness against **overfitting on out-of-time data**.

Software Engineering Intern | ***Infosys – Renault**, Paris* *Mar 2023 – Oct 2023*
- Built backend modules for supply-chain platforms in **Java** and **PostgreSQL**.

Quantitative Research & Engineering Projects

Crypto Perpetuals Alpha Strategy | *ML for Financial Markets, HEC Paris* *2026*
- Ranked **#1 of 13 teams (˜50 students)** on out-of-sample Sharpe for a dollar-neutral long-short strategy across **411 cryptocurrency perpetual futures** (gross of transaction costs).
- Engineered **379** candidate features — **316** retained after **Spearman IC** filtering and correlation pruning — into a **Ridge + XGBoost** ensemble under a rolling 12-month **walk-forward**, with PCA, scaler, and models refit per window and the IC filter fit once on train and frozen as an explicit anti-leakage choice.

Leukemia Survival Prediction — QRT / ENS Data Challenge | *Cox PH, Python* *2025*
- Predicted overall survival on clinical + genomic data from 24 hospitals (**4,500+ patients**) using a **Cox Proportional Hazards** model; finished **top 12%**.

Cognitive Alpha — Spatial Decision Analytics, FIFA World Cup 2022 | *Personal Project, Python* *2025–2026*
- Quantified, for each open-play pass, the gap between the player's decision and the spatially optimal alternative: trained an **Expected Threat (xT)** model via **Markov-chain value iteration** on **126k possession actions**, with a NumPy-vectorised pitch-control engine and a **Streamlit** dashboard over ∼5.4 GB of tracking data.

Technical Skills & Activities

Programming: **Python** (primary), SQL (PostgreSQL), R, MATLAB, Bash; C++ / Java (working knowledge)
Quantitative & ML: Probability & Statistics, Time-Series, Survival Analysis, Signal/Alpha Research
Tools: PyTorch, XGBoost, Scikit-learn, NumPy, Pandas, MLflow, Docker, Git, Kafka, FastAPI, Dataiku, Linux
Self-study & Certs: Stochastic calculus & options theory (Hull, Natenberg, Wilmott); Akuna Capital Options 101 & 202; **Jane Street** puzzles
Languages: French (Native), English (Fluent); Dutch, Spanish, Portuguese & Arabic (B2)
Activities: Built and sold a **200k-follower** X/Twitter account to **Polymarket**; private mathematics tutor (3+ years); **Chess** (1700 Elo)